Where Refusal Lives: Refusal Is Geometrically Separate from Moral Representation Across Model Families, but Not Always from Moral Behavior

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Abstract

A refusal direction can be stripped from many instruction-tuned models by a single rank-one weight edit. A prior paper showed that in OLMo-3 this direction is geometrically separate from the model's moral representation: it is almost entirely residual against the moral-foundations subspace, and ablating it removes refusal while leaving moral comprehension intact. We ask whether that separation is a property of language models in general or an artifact of one model's post-training. Holding the extraction conventions fixed, we run the same decomposition across three dense base+instruct families of similar size, OLMo-3 7B, Qwen2.5-7B, and Llama-3.1-8B. Two results separate cleanly. Representationally the dissociation is family-invariant: in all three the refusal direction is $\sim$99% residual and near-orthogonal to the six-foundation moral subspace (projection fraction 0.07–0.13), and ablating it leaves the linear moral representation untouched (fresh probe accuracy 1.0, effective dimensionality 5). Behaviorally the picture is family-dependent: in OLMo-3 and Qwen2.5 a single-direction ablation removes refusal and leaves behavioral moral judgment intact, the clean comprehension/compliance dissociation, but in Llama-3.1 the refusal direction is both harder to ablate and *entangled* with moral judgment. Removing it degrades Llama's behavioral moral judgment dose-dependently and refusal-direction-specifically: a magnitude-matched random weight perturbation, even at twice the magnitude, leaves judgment unchanged, while the matching perturbation along the refusal direction degrades it, all while the linear moral representation stays perfectly preserved. The refusal/moral separation that holds representationally across families therefore does not always hold behaviorally. We treat Llama's coupling as a measured property of one released model, established through controls rather than asserted, and make no claim about its training recipe. The work is diagnostic and produces no technique for constructing harder-to-remove refusal.

1 Introduction

A refusal direction can be removed from many instruction-tuned models by a single rank-one weight edit. Arditi et al. (Arditi et al., 2024) showed that refusal behavior is mediated by one linear direction in the residual stream, and that orthogonalizing the model's weights against it suppresses refusal across a wide range of harmful requests; the Heretic tool (p-e-w, 2025) automates the same operation across Qwen, Llama, and other families. That refusal is single-direction-ablatable is, by now, well established and is not what we study.

The open question this leaves is different. Two models can both have a refusal direction that a rank-one edit removes, yet differ in *what that direction is made of*. In one model the ablatable direction might be built largely out of moral features, so that removing it also disturbs the model's moral

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representation; in another it might be nearly orthogonal to moral structure, a separate mechanism that ablation can excise cleanly. Heretic does not distinguish these cases, because it measures only whether refusal drops, not where the refusal direction sits relative to the model's moral geometry. **Is the ablatable refusal direction morally grounded, or geometrically separate from moral structure?** That routing question is what we test.

A prior paper in this series answered it for one model. On OLMo-3 7B, the refusal direction is almost entirely residual against the six-foundation moral subspace: its projection onto that subspace is small, its cosine to each Moral Foundation direction is near zero, and ablating it removes refusal while leaving moral comprehension and behavioral moral judgment intact (Reblitz-Richardson, 2026a). Comprehension and compliance are separable, and refusal is the separable part. The natural worry is that OLMo-3 is a weak test. Its post-training is lighter than that of the most heavily aligned open models, and a model whose refusal is shallow might show a non-moral refusal direction for that reason alone. If so, the dissociation would be a property of OLMo-3, not of language models, and the interesting question, whether refusal is ever morally grounded, would have to be asked on a more strongly aligned model.

This paper runs the same decomposition across three model families to find out. We hold the extraction conventions fixed and compare a dense $\sim$7–8B base+instruct pair from each of three families: OLMo-3 7B (the anchor), Qwen2.5-7B (Qwen Team, 2025), and Llama-3.1-8B (Llama Team, AI @ Meta, 2024). For each we extract the refusal direction (Arditi/Heretic), the six-foundation moral subspace, and a persona direction, and decompose refusal into a moral part, a persona part, and a residual. We then apply the single-direction ablation and re-measure moral comprehension, both the linear moral representation and behavioral moral judgment. The panel is deliberately narrow: dense (not mixture-of-experts, whose output dilution is a known confound (Reblitz-Richardson, 2026c)), size-matched, and non-reasoning, so that "OLMo looks different" cannot be an artifact of architecture, scale, or a thinking-mode trace.

Two results separate, and the separation is the paper.

Finding 1: The representational dissociation is family-invariant. In all three families the refusal direction is $\sim 99\%$ residual and near-orthogonal to the moral subspace (projection fraction 0.07–0.13, mean absolute cosine to the foundations 0.04–0.075), with essentially no persona component. Ablating it leaves the *linear moral representation* untouched: fresh per-foundation probe accuracy stays at 1.0 and the moral subspace keeps its five effective dimensions, in every family. Whatever else differs across these models, none of them stores refusal inside its moral geometry, and removing refusal does not damage what the model linearly encodes about morality. OLMo-3 is representative, not an outlier.

Finding 2: The behavioral dissociation is clean where refusal is cleanly removable. In OLMo-3 and Qwen2.5 the single-direction ablation drives refusal to zero (compliance on a persona-shift battery rises to 1.0) and leaves behavioral moral judgment intact (within two points of the unablated model). This is the comprehension/compliance dissociation realized behaviorally: the model still judges moral scenarios as before while refusing nothing, exactly the high-comprehension / low-compliance cell.

Finding 3: Llama-3.1 is the exception, on two counts. First, its refusal is markedly harder to remove: a single-direction ablation that fully strips OLMo and Qwen only partially strips Llama, and the mid-depth layers that work for the others do nothing for it. Second, and more striking, ablating Llama's refusal direction degrades its behavioral moral judgment while leaving the linear moral representation perfectly intact. We establish that this degradation is *refusal-direction-specific* and *dose-dependent* rather than a side effect of perturbing the weights: a magnitude-matched random weight perturbation, even at twice the magnitude, leaves moral judgment unchanged, while the matching perturbation along the refusal direction degrades it, and the degradation grows monotonically with the fraction of the refusal direction removed. In Llama-3.1, refusal and behavioral moral judgment are *entangled*. We report this as a property of one released model, measured through controls, and use it to motivate where the question should go next, not as a cross-model law.

Taken together: the refusal/moral separation that holds *representationally* across families does not always hold *behaviorally*. The thin-refusal worry is refuted, because the dissociation reproduces even in better-aligned models; but the behavioral side is family-dependent, and the one place it breaks down, Llama-3.1, shows refusal entangled with moral behavior even though the moral representation is intact and family-invariant. A bonus observation follows from the ablation step: single-direction refusal removability is itself family-dependent, and the family where refusal is hardest to remove is also the one where it is morally entangled. With $n = 1$ on that co-occurrence we do not draw a correlation; we read it as a signal that the models worth probing next are the more strongly aligned and reasoning-capable ones, where both properties may be more common.

This work continues a series on moral representation in language models: that moral content is linearly decodable early in pre-training (Reblitz-Richardson, 2026d), encoded uniformly across mixture-of-experts experts (Reblitz-Richardson, 2026c), organized into a structured, five-dimensional framework geometry (Reblitz-Richardson, 2026e) whose directions are causal for moral judgments (Reblitz-Richardson, 2026b), and coupled to behavior during post-training as a separable compliance mechanism (Reblitz-Richardson, 2026a). The present paper asks whether that last result, the separability of refusal from moral representation, generalizes, and finds that it does representationally and usually but not always behaviorally.

Everything here is diagnostic. We characterize where refusal sits relative to moral structure and what ablating it does to moral comprehension, using random and feature-matched controls to attribute the effects we find. We produce no method for making refusal harder to remove, and we make no claim about any model's training data or recipe, which we do not have.

2 Related Work

The refusal direction and abliteration. Arditi et al. (Arditi et al., 2024) showed that refusal in instruction-tuned models is mediated by a single linear direction in the residual stream: the difference in mean activations between harmful and harmless instructions, extracted at one layer, suppresses refusal when projected out of the model's residual-writing weights and induces refusal when added. The Heretic tool (p-e-w, 2025) automates this "abliteration" across model families with a black-box search over layer and strength. This line of work establishes that refusal is, behaviorally, single-direction-removable; it does not ask where that direction sits relative to a model's moral representation, which is our question.

Moral foundations and moral probing. Moral Foundations Theory (Graham et al., 2013, Haidt, 2012) decomposes moral cognition into a small number of foundations (care, fairness, loyalty, authority, sanctity, liberty). Earlier papers in this series use linear probes (Alain and Bengio, 2017, Belinkov, 2022) to recover these foundations from model activations and to characterize their geometry. We reuse that probing machinery to define a six-foundation moral subspace and to measure whether the linear moral representation survives refusal ablation.

Linear features and concept directions. A growing body of work treats high-level concepts as linear directions in representation space (Bolukbasi et al., 2016, Park et al., 2024), manipulable by addition and ablation (Zou et al., 2023) and removable by concept erasure (Belrose et al., 2023). Refusal (Arditi et al., 2024), persona (Wang et al., 2025), and the assistant's default identity (Lu et al., 2026) have all been described as such directions. Our decomposition asks how the refusal direction relates to the moral and persona directions in the same model, and our ablation controls ask whether a behavioral effect is specific to the refusal direction or generic to weight perturbation.

Persona and the assistant axis. Wang et al. (Wang et al., 2025) identify persona features that control emergent misalignment, and Lu et al. (Lu et al., 2026) characterize an "assistant axis" that stabilizes the model's default persona. The prior paper in this series found OLMo-3's refusal direction nearly orthogonal to both the moral subspace and the persona direction; here we carry the persona direction through as a structured, non-moral control when attributing the Llama coupling.

This series. Across pre-training, moral content is linearly decodable early and across most layers (Reblitz-Richardson, 2026d), encoded uniformly across mixture-of-experts experts rather than specialized (Reblitz-Richardson, 2026c), and organized into a five-dimensional framework geometry (Reblitz-Richardson, 2026e) whose directions are causal for moral judgments (Reblitz-Richardson, 2026b). The most recent paper (Reblitz-Richardson, 2026a) showed, on OLMo-3, that post-training couples a separable refusal mechanism to this pre-existing moral representation rather than teaching morality anew, and that the refusal direction is geometrically separate from moral structure. The present paper tests whether that separability is a property of language models in general.

3 Methodology

3.1 Model panel

We compare a dense base+instruct pair from each of three families: OLMo-3 7B (base `Olmo-3-1025-7B`, instruct `Olmo-3-7B-Instruct`) as the anchor that reproduces the prior single-model result, Qwen2.5-7B (Qwen Team, 2025), and Llama-3.1-8B (Llama Team, AI @ Meta, 2024). The panel is fixed by three controls, each motivated by a known confound. It is **dense only**: a previous paper found that mixture-of-experts blocks dilute the moral signal in the residual stream by a large factor (Reblitz-Richardson, 2026c), so comparing a dense anchor against a sparse model would confound routing geometry with dilution. It is **size-matched** at $\sim$7–8B, because moral-representation properties are scale-dependent (Reblitz-Richardson, 2026d) and we do not want scale as a free variable. And it is **non-reasoning**: we deliberately use Qwen2.5 and Llama-3.1 rather than their reasoning-mode successors, because refusal emitted inside a thinking trace is a different object than refusal in a direct response, and OLMo-3 has no such mode. Whether these results hold in reasoning models is a separate question we return to in the discussion.

3.2 Identical extraction conventions

The validity of any cross-model comparison rests on applying identical conventions to every model; otherwise "model A looks different" can be a measurement artifact. We extract every direction with the same pooling (mean over content tokens), the same direction estimator (mean-difference, with a seeded linear probe reported alongside), the same contrast sets, and the same input format per direction kind. Moral and persona directions are extracted on the **base** model from raw text; the refusal direction is extracted on the **instruct** model in chat format. The moral and persona collection path pools raw text directly and never applies a chat template, so the fact that Qwen2.5's base tokenizer ships a chat template does not leak into the geometry.

Layer indices are the only thing that legitimately differs across models, and we fix them by depth fraction rather than absolute index. The anchor's stable band is layers 15–31 of 32 (depth fractions 0.47–0.97) and its headline layer is the depth-0.5 layer (16 of 32). Mapping these fractions through $\mathrm{round}(f \cdot n_{\text{layers}})$ gives band $[15, 31]$ / layer 16 for OLMo-3 (32 layers), band $[13, 27]$ / layer 14 for Qwen2.5 (28 layers), and band $[15, 31]$ / layer 16 for Llama-3.1 (32 layers). We report the decomposition at the depth-0.5 headline layer and summarize the stable band as a mean. Because OLMo-3 is the anchor, this rule reproduces the prior paper's numbers exactly: the OLMo refusal projection-fraction we recover, 0.104, matches the published 0.1044, confirming the conventions transferred before any comparison is read.

3.3 Direction extraction

For each model we extract three sets of directions. The **moral subspace** is the span of six per-foundation probe-weight directions, fit on a balanced declarative moral/neutral probing set and orthonormalized by SVD into a five- to six-dimensional basis S. The **persona direction** is a single probe direction separating personal from non-personal identity statements. The **refusal direction** r is the Arditi/Heretic estimator: the unit difference between the mean last-token residual activations of harmful and harmless instructions (the real Arditi prompt set, 400 of each, in chat format), computed per layer.

3.4 Refusal decomposition

We decompose the unit refusal direction r at a layer into three energy (squared-norm) fractions that partition it exactly. With S the orthonormal moral subspace basis and u the unit persona component orthogonal to S,

$$\text{mft\_frac} = \|\mathrm{proj}_S(r)\|^2, \quad \text{persona\_frac} = (r \cdot u)^2, \quad \text{residual\_frac} = 1 - \text{mft\_frac} - \text{persona\_frac}.$$

Because $S \perp u$ by construction the three sum to 1. We also report the norm-ratio $\sqrt{\text{mft\_frac}}$ (the projection fraction of the prior paper), the cosine of r to each foundation, and the cosine of r to the persona direction.

3.5 Refusal consolidation

To check whether any model's low moral fraction is an artifact of a diffuse refusal rather than genuine separation, we measure how concentrated the refusal signal is. At the headline layer we report the separation margin d' and the single-direction AUC of harmful vs. harmless activations projected onto r. We then ask whether the single ablatable direction captures *all* the linear separability or whether residual multi-direction refusal exists, by comparing, on held-out prompts, the AUC of the mean-difference direction against the AUC of a regularized full-rank classifier. The full classifier is Ledoit-Wolf shrinkage linear discriminant analysis: at hidden dimension $\gg$ prompt count an unregularized classifier overfits and reports an inflated AUC, while data-adaptive shrinkage avoids that, and its $\rho \to 1$ limit is exactly the mean-difference direction, so the single-vs-full AUC gap is interpretable. Across the stable band we report the effective rank of the per-layer refusal directions, computed uncentered (centering a near-collinear set subtracts its shared component and reports full rank, which would misread a consolidated refusal as diffuse).

3.6 Ablation and the comprehension battery

We apply the Arditi/Heretic single-direction ablation, orthogonalizing every residual-writing matrix (attention `o_proj`, MLP `down_proj`) against r at the chosen layer, $W \leftarrow W - r\,r^{\top}W$. Because the depth-$0.5$ ablation that fully strips OLMo and Qwen only partially strips Llama, we **sweep the ablation layer by depth fraction** (0.4–0.8) for every model and ablate at the layer that most reduces the harmful refusal rate, so the comprehension battery runs at a layer where the ablation actually takes; we report the chosen layer and the refusal removed per model.

On the unablated instruct model and on the ablated model we run the same battery: fresh per-foundation probe accuracy and the effective dimensionality of the moral subspace (the linear *representation*), behavioral moral-judgment accuracy on a 48-scenario moral-foundations probe and persona-shift compliance (the *behavior*), and a moral-text dependency score. The discriminator is the instruct→ablated **delta** on each comprehension measure: a delta near zero is a clean dissociation (refusal is not load-bearing for comprehension); a large delta means ablating refusal degrades comprehension in that model.

3.7 Controls for the Llama effect

Where ablation degrades behavioral moral judgment, two controls separate genuine coupling from confounds.

Random-direction control. We ask whether the judgment drop is specific to the refusal direction or generic to a weight perturbation of that magnitude. We record the per-matrix Frobenius norm removed by the refusal ablation and then perturb the same matrices with random Gaussian noise of *identical per-matrix norm*, repeated over several independent draws. A plain random unit direction is near-orthogonal to the weights and removes almost nothing, which would bias the test toward "specific"; matching the magnitude is the fair null. We also ablate the persona direction, a real, decodable, comparable-magnitude, non-refusal/non-moral feature, as a structured control. For every condition we measure both moral judgment and the linear representation.

Ablation-strength sweep. We vary the fraction α of the refusal direction removed, $W \leftarrow W - \alpha\,r\,r^{\top}W$, from partial ($\alpha < 1$) through full ($\alpha = 1$) to over-ablation ($\alpha > 1$, a measurement probe used to reach fuller refusal removal, not a deployable intervention), and at each step measure how much refusal is actually stripped (harmful refusal rate), moral judgment with a paired per-scenario bootstrap confidence interval, and the linear representation, against a magnitude-matched random null at the same removal magnitudes. If judgment degrades monotonically as more refusal is removed, and a matched random perturbation leaves it intact, the coupling is graded and direction-specific; if judgment is flat then cliffs at one step while refusal removal saturates, the drop is a destabilization artifact. Effect sizes are anchored on the coherent $\alpha \leq 1$ regime; the over-ablation regime is reported only as a caveated endpoint, because at the strongest perturbations generation degrades and moral judgment falls below chance.

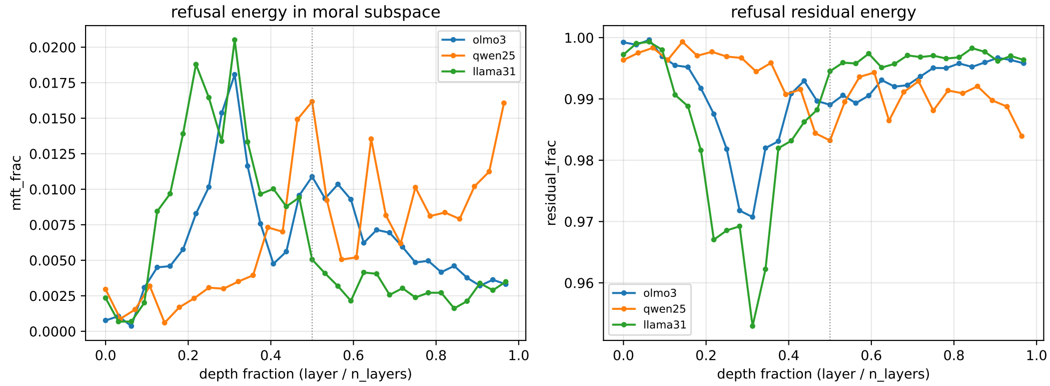

Figure 1: Refusal-direction energy in the moral subspace (left) and residual (right) as a function of depth fraction, for all three families. The three curves track together: across depth the refusal direction places almost none of its energy in the moral subspace and is overwhelmingly residual. The dotted line marks the depth-0.5 headline layer.

4 Results

4.1 Refusal is residual and orthogonal to morality in every family

Decomposing each model's refusal direction at the depth-0.5 headline layer gives the same shape in all three families (Table 1). Refusal is 98–99% residual energy, its projection onto the six-foundation moral subspace is small (norm-ratio 0.07–0.13, energy fraction 0.005–0.016), its mean absolute cosine to the foundations is 0.04–0.075, and its persona component is essentially zero (cosine to the persona direction 0.02–0.05). The OLMo-3 value reproduces the prior single-model paper exactly: a projection fraction of 0.104 against the published 0.1044, which confirms the conventions transferred before any cross-model claim is read. Figure 1 shows the pattern holds across depth, not just at one layer.

Table 1: Refusal-direction decomposition at the depth-0.5 headline layer. No family routes refusal materially through moral features; even the largest moral projection (Qwen, 0.127) is the same regime as the anchor (0.104).

metric (headline layer)	OLMo-3	Qwen2.5	Llama-3.1
residual energy fraction	0.989	0.983	0.995
moral energy fraction	0.011	0.016	0.005
moral projection fraction	0.104	0.127	0.071
mean abs. cosine to foundations	0.060	0.075	0.041
persona energy fraction	0.000	0.001	0.000
single-direction AUC	1.00	1.00	1.00
single-vs-full-rank AUC gap	0.000	0.000	0.000
across-band refusal eff. rank	3	6	4

No family has a materially more moral refusal direction than the anchor. The spread in projection fraction is narrow (0.07–0.13), and the model with the *most* moral projection (Qwen) is also the one with the *most* diffuse refusal (effective rank 6 across the band, against 3 for OLMo), the opposite of what a "diffuseness explains the low moral fraction" account would predict. The single-direction AUC is 1.0 and the single-vs-full-rank gap is 0.000 in every family: the one ablatable direction captures all the linear separability of harmful from harmless, with no residual multi-direction refusal hiding behind it. The thin-refusal worry is refuted: the geometric separation of refusal from morality is not an OLMo artifact but a family-invariant property.

4.2 Ablating refusal preserves the linear moral representation everywhere

Single-direction refusal removability is itself family-dependent. Sweeping the ablation layer by depth fraction, OLMo's harmful refusal rate falls from 0.575 to 0.000 and Qwen's from 1.000 to 0.000, but Llama's only falls from 0.900 to 0.475, and the mid-depth layers that strip the others do nothing for Llama; only a shallow layer reduces its refusal, and only partially (Table 2).

At each model's best ablation layer, ablating refusal leaves the *linear moral representation* untouched in all three families: fresh per-foundation probe accuracy stays at 1.0 and the moral subspace keeps its five effective dimensions, before and after (Table 2). Whatever refusal removal does to a model, it does not damage what the model linearly encodes about morality. This is the representational dissociation, and it is family-invariant.

Table 2: Ablation and the comprehension battery at each model's swept ablation layer. Representation is preserved everywhere; behavioral moral judgment is preserved in OLMo and Qwen but drops in Llama, the only model whose refusal does not fully strip.

metric	OLMo-3	Qwen2.5	Llama-3.1
ablation layer (depth)	19 (0.59)	14 (0.50)	13 (0.41)
refusal rate: clean → ablated	0.575 → 0.000	1.000 → 0.000	0.900 → 0.475
probe accuracy: clean → ablated	1.0 → 1.0	1.0 → 1.0	1.0 → 1.0
eff. dimensionality: clean → ablated	5 → 5	5 → 5	5 → 5
persona-shift compliance: clean → ablated	0.75 → 1.00	0.90 → 1.00	0.70 → 0.95
moral judgment: clean → ablated	0.75 → 0.79	0.875 → 0.812	0.75 → 0.604

4.3 The behavioral dissociation is clean in OLMo-3 and Qwen2.5

In the two families where the single-direction ablation fully removes refusal, it also leaves *behavioral* moral judgment intact. OLMo's persona-shift compliance rises from 0.75 to 1.00 (refusal stripped, every persona gap closing to zero) while its moral-judgment accuracy is essentially flat, $0.75 \to 0.79$. Qwen's compliance rises from 0.90 to 1.00 and its judgment moves from 0.875 to 0.812. This is the comprehension/compliance dissociation realized end to end: the model judges moral scenarios as before while refusing nothing. For these families the prior single-model result generalizes without qualification, in both its representational and behavioral form.

4.4 Llama-3.1: refusal entangled with moral judgment

Llama behaves differently, and the difference is the result. At its best ablation layer its persona-shift compliance does rise ($0.70 \to 0.95$), but its moral-judgment accuracy drops, $0.75 \to 0.604$. The linear representation is untouched (probe accuracy 1.0, effective dimensionality 5), so this is a behavioral effect, not a representational collapse. Two controls establish that the drop is specific to the refusal direction and graded in the amount of refusal removed.

The drop is refusal-direction-specific. Perturbing the same weight matrices with random Gaussian noise of *identical per-matrix Frobenius norm* to the refusal ablation leaves moral judgment unchanged: over eight independent draws the matched-random judgment is 0.747 ± 0.007, essentially the clean value of 0.75, against the refusal-ablated 0.604 (a -21σ outlier below the matched-random null). Ablating the persona direction, a real, decodable, comparable-magnitude, non-refusal feature, also leaves judgment at 0.75. Only ablating the refusal direction degrades behavioral moral judgment, and the linear representation stays at probe accuracy 1.0 and effective dimensionality 5 under every one of these conditions. So Llama is not generally fragile to weight perturbation of this magnitude, and it is not the case that ablating any salient feature degrades judgment.

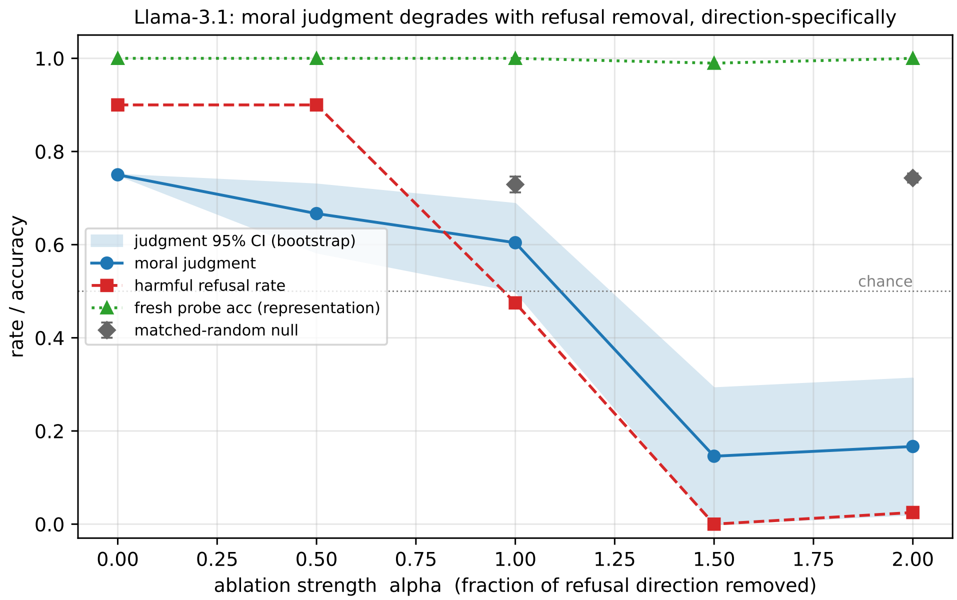

Figure 2: Llama-3.1 ablation-strength sweep. As the fraction α of the refusal direction removed grows, behavioral moral judgment (blue, with bootstrap CI) falls while the linear moral representation (green, fresh probe accuracy) stays at 1.0 and a magnitude-matched random perturbation (grey diamonds, even at α=2) leaves judgment ≈ 0.74. The harmful refusal rate (red) drops only once over-ablation ($\alpha \geq 1.5$) is reached. Effect sizes are read from the coherent $\alpha \leq 1$ regime; the $\alpha \geq 1.5$ collapse below chance reflects degraded generation and is reported only as an endpoint.

The drop is dose-dependent. Figure 2 sweeps the fraction α of the refusal direction removed. Moral judgment falls monotonically as more refusal is removed (Spearman correlation between refusal removed and judgment drop $= 1.0$), and within the coherent regime the partial-ablation drops are individually significant: at $\alpha = 0.5$ the drop is 0.083 (95% bootstrap CI $[0.02, 0.17]$, $n = 48$ scenarios) and at the full single-direction strength $\alpha = 1$ it is 0.146 (CI $[0.06, 0.25]$). At $\alpha = 0.5$ the perturbation along the refusal direction already costs judgment even though it has not yet reduced behavioral refusal (refusal rate still 0.90), the signature of an entanglement at the level of the direction rather than of the refusal behavior it controls. Throughout, the magnitude-matched random null stays flat, 0.74 even at twice the full-ablation magnitude, so the dose-response cannot be a magnitude artifact.

We anchor the coupling on this $\alpha \leq 1$ regime, where the model still generates coherently. Driving α higher does eventually strip Llama's refusal in full ($\alpha = 1.5$ takes the refusal rate to 0.0) and collapses moral judgment far below chance, but at those over-ablation strengths the model's generation is degraded, so we report that collapse only as a caveated endpoint and never as the headline effect. The supported claim is the coherent one: in Llama-3.1, refusal and behavioral moral judgment are entangled, the entanglement is specific to the refusal direction, and it grows with the amount of refusal removed.

Finally, the two ways Llama differs, its refusal being the hardest to remove and its refusal being morally entangled, co-occur in this single model. We do not draw a correlation from one model; we note the co-occurrence and treat it as a question for the next study rather than a finding of this one.

5 Discussion

A representational property generalizes; a behavioral one does not. The cleanest way to state the result is to separate two things that "comprehension" can mean. The *linear moral representation*, what a probe can decode from activations and how those foundation directions are arranged, is untouched by refusal ablation in all three families, and the refusal direction is residual and orthogonal to it in all three. That property is family-invariant, and it settles the question the paper set out to ask:

the geometric separation of refusal from morality reported for OLMo-3 is not an artifact of a weakly aligned model. But *behavioral moral judgment*, what the model actually outputs when asked to judge a scenario, comes apart from the representation in Llama-3.1: there, removing the refusal direction degrades the judgment while leaving the representation intact. The two senses of comprehension travel together in OLMo and Qwen and separate in Llama, which is worth keeping distinct in how the field talks about "comprehension surviving ablation."

Refusal removability is not universal. A side result of the ablation step is that the single-direction, single-edit removability that motivates abliteration tooling is family-dependent in degree. OLMo and Qwen lose refusal completely under the uniform orthogonalization; Llama, under the same protocol applied at its own best layer, loses only half, and the mid-depth layers that work for the others have no effect on it. We do not claim Llama's refusal is un-ablatable, only that the simplest single-direction edit is markedly less effective on it. A more elaborate search of the kind Heretic performs may well do better; characterizing that is outside our scope and, by design, not something we pursue, since it would amount to engineering more effective refusal removal.

The Llama coupling, stated precisely. What we have shown about Llama is that ablating its refusal direction degrades behavioral moral judgment, that this degradation is specific to the refusal direction (a magnitude-matched random perturbation, and an ablation of the comparable-magnitude persona direction, do not produce it), and that it grows with the fraction of refusal removed. We call this an *entanglement* between refusal and behavioral moral judgment that is *dose-dependent* and *refusal-direction-specific*. We deliberately do not say that refusal "routes through" moral machinery or shares a circuit with it: a directional ablation experiment licenses claims about which direction, perturbed by how much, changes which behavior, and not claims about internal mechanism or shared structure. The honest summary is a dose-response between a direction and a behavior, with the representation held fixed throughout.

Why this matters for where to look next. In this one model the two unusual properties, a refusal direction that resists single-direction removal and a refusal direction entangled with moral behavior, occur together. With a single model we cannot tell whether that co-occurrence is meaningful or coincidental, and we are careful not to present it as a cross-model relationship. What it does is locate the interesting regime. If harder-to-remove and morally entangled refusal tend to appear together, they should appear more often in the models with the strongest alignment and, plausibly, in reasoning models, where refusal is produced through a deliberative trace rather than a direct response. Those are precisely the models we excluded here to keep the comparison controlled. Extending the panel to reasoning models is the natural next study, and it is where an $n > 1$ test of the co-occurrence belongs.

Limits. The co-occurrence of ablation-resistance and moral entanglement rests on a single model. The panel is dense, $\sim$7–8B, and non-reasoning by design, so the results speak to that regime and not yet to larger, sparser, or reasoning-capable models. The moral-judgment probe is 48 scenarios; we report bootstrap confidence intervals so the effect size is not read off a handful of items, but a larger behavioral battery would tighten it. Over-ablation degrades generation, so the strongest perturbations are not clean measurements of moral judgment and we use them only as a directional endpoint. And throughout we measure where refusal sits and what perturbing it does; we do not have any model's training data or recipe and make no claim about why a given model's refusal is shaped the way it is.

Safety. Every measurement here is diagnostic. We characterize an existing property of released models, the geometry of their refusal directions and what ablating those directions does to moral comprehension, using random and feature-matched controls to attribute the effects. Llama's coupling is a measured property of one released model, not a technique we built; the over-ablation sweep is a measurement probe, not an intervention; and we produce no method for making refusal harder to remove. The reverse question, whether refusal *should* be more morally grounded and how one might achieve that during pre-training rather than post-training, is raised by these results but is a separate program, and not one we advance here.

6 Conclusion

We asked whether the geometric separation of refusal from moral representation, reported for a single model, is a property of language models in general or an artifact of one model's post-training. Running the same decomposition across three dense base+instruct families under identical conventions, we find that representationally the separation is family-invariant: in OLMo-3, Qwen2.5, and Llama-3.1

the refusal direction is $\sim 99\%$ residual against the moral subspace and near-orthogonal to it, and ablating it leaves the linear moral representation untouched. Behaviorally, the picture is family-dependent. In OLMo-3 and Qwen2.5 a single-direction ablation removes refusal and leaves behavioral moral judgment intact, the clean comprehension/compliance dissociation. In Llama-3.1, refusal is both harder to remove and entangled with moral judgment: ablating it degrades the judgment dose-dependently and refusal-direction-specifically, verified against a magnitude-matched random null and a persona-direction control, while the linear moral representation stays intact.

The separation of refusal from morality that holds representationally across families therefore does not always hold behaviorally, and the single model where it breaks down is also the one whose refusal is hardest to remove. We report that co-occurrence as an $n = 1$ observation, not a law, and read it as a pointer toward the more strongly aligned and reasoning-capable models that a controlled panel had to exclude. The work is diagnostic throughout: it characterizes where refusal sits relative to moral structure and what removing it does to moral comprehension, and builds no technique for making refusal harder to remove.

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Appendices

Supplementary material.

A Full Tables

A.1 Refusal-direction decomposition and consolidation

All values at the depth-0.5 headline layer except the band columns, which are means over the stable band (OLMo/Llama layers 15–31, Qwen 13–27). Energy fractions partition the unit refusal direction and sum to 1. The norm-ratio is $\sqrt{\text{moral fraction}}$, the projection fraction reported by the prior single-model paper. The single-vs-full-rank AUC gap compares the mean-difference direction against a Ledoit-Wolf shrinkage LDA on held-out prompts; the across-band effective rank and participation ratio are computed uncentered.

Table 3: Full refusal-direction decomposition and consolidation per family.

metric	OLMo-3	Qwen2.5	Llama-3.1
moral energy fraction	0.0109	0.0162	0.0051
moral norm-ratio (projection fraction)	0.1043	0.1272	0.0711
residual energy fraction	0.9890	0.9832	0.9945
persona-unique energy fraction	0.0001	0.0006	0.0004
mean abs. cosine to foundations	0.0605	0.0746	0.0407
cosine(refusal, persona)	0.0241	0.0499	0.0301
band mean moral fraction	0.0064	0.0100	0.0035
band mean residual fraction	0.9933	0.9894	0.9960
margin d' (harmful vs. harmless)	9.81	5.57	7.62
single-direction AUC	0.9999	0.9998	0.9995
single-vs-full-rank AUC gap	0.0001	0.0000	0.0002
across-band effective rank	3	6	4
across-band participation ratio	1.93	3.16	2.23

A.2 Ablation and comprehension battery (swept ablation layer)

Ablation at each model's swept (max-refusal-reduction) layer. Probe accuracy and effective dimensionality measure the linear representation; moral judgment and persona-shift compliance measure behavior. The moral-text dependency score is a sensitive cross-entropy difference on 240 texts and is noisy at this sample size; we include it for completeness but do not draw conclusions from it.

Table 4: Comprehension battery, instruct vs. ablated, per family.

metric	OLMo-3	Qwen2.5	Llama-3.1
ablation layer (depth fraction)	19 (0.59)	14 (0.50)	13 (0.41)
harmful refusal rate: clean	0.575	1.000	0.900
harmful refusal rate: ablated	0.000	0.000	0.475
fresh probe accuracy: clean $\to$ ablated	$1.0 \to 1.0$	$1.0 \to 1.0$	$1.0 \to 1.0$
effective dimensionality: clean $\to$ ablated	$5 \to 5$	$5 \to 5$	$5 \to 5$

metric	OLMo-3	Qwen2.5	Llama-3.1
persona-shift compliance: clean $\rightarrow$ ablated	0.75 $\rightarrow$ 1.00	0.90 $\rightarrow$ 1.00	0.70 $\rightarrow$ 0.95
moral judgment: clean $\rightarrow$ ablated	0.750 $\rightarrow$ 0.792	0.875 $\rightarrow$ 0.812	0.750 $\rightarrow$ 0.604
moral dependency: clean $\rightarrow$ ablated	0.064 $\rightarrow$ 0.049	0.081 $\rightarrow$ 0.000	0.052 $\rightarrow$ 0.071

A.3 Random-direction control (Llama-3.1, ablation layer 13)

Each condition ablates/perturbs and re-measures both moral judgment and the linear representation. The matched-random null perturbs the same residual-writing matrices with Gaussian noise of per-matrix Frobenius norm identical to the refusal ablation's (total removed $\|W\|_F = 10.23$), over eight draws. Persona ablates the comparable-magnitude persona direction.

Table 5: Random-direction and persona controls. Refusal judgment is a -20.8σ outlier below the matched-random null; representation is preserved under every condition.

condition	moral judgment	probe accuracy	eff. dimensionality
clean (no perturbation)	0.750	1.0	5
refusal-direction ablation	0.604	1.0	5
persona-direction ablation	0.750	1.0	5
matched-random null (mean $\pm$ sd, n=8)	0.747 $\pm$ 0.007	1.0	5

A.4 Ablation-strength sweep (Llama-3.1, ablation layer 13)

The fraction α of the refusal direction removed varies from partial through full to over-ablation. Moral-judgment drops are paired per-scenario bootstrap estimates against the α=0 baseline (5000 resamples, n=48 scenarios). The matched-random null at α=1.0 is 0.729 ± 0.017 and at α=2.0 is 0.743 ± 0.010 (n=3 draws each). Effect sizes are read from the coherent $\alpha \leq 1$ regime; at $\alpha \geq 1.5$ moral judgment falls below chance as generation degrades.

Table 6: Refusal-ablation strength sweep. Spearman correlation between refusal removed and judgment drop $= 1.0$.

α	refusal rate	refusal removed	moral judgment	judgment drop [95% CI]	probe acc	eff. dim
0.0	0.900	0.000	0.750	—	1.0	5
0.5	0.900	0.000	0.667	0.083 [0.02, 0.17]	1.0	5
1.0	0.475	0.425	0.604	0.146 [0.06, 0.25]	1.0	5
1.5	0.000	0.900	0.146	0.604 [0.46, 0.75]	0.99	5
2.0	0.025	0.875	0.167	0.583 [0.44, 0.73]	1.0	5

B Reproducibility

Models. All six checkpoints are public Hugging Face releases: OLMo-3 7B (`allenai/Olmo-3-1025-7B`, `allenai/Olmo-3-7B-Instruct`) (Ai2, 2025), Qwen2.5-7B (`Qwen/Qwen2.5-7B`, Qwen/Qwen2.5-7B-Instruct) (Qwen Team, 2025), and Llama-3.1-8B (meta-llama/Llama-3.1-8B, meta-llama/Llama-3.1-8B-Instruct) (Llama Team, AI @ Meta, 2024). Layer counts are 32, 28, and 32; hidden sizes 4096, 3584, and 4096. Models load in `float16`.

Conventions. Layer indices are set by depth fraction: the stable band is the OLMo anchor's $[15, 31]/32$, depth fractions $(0.469, 0.969)$, mapped by $\mathrm{round}(f \cdot n_{\text{layers}})$ to $[15, 31]$ (OLMo, Llama) and $[13, 27]$ (Qwen); the headline layer is depth-0.5 (16, 14, 16). Moral and persona directions are extracted on the base model from raw text; the refusal direction is extracted on the instruct model in chat format. Direction estimator is mean-difference for the refusal direction and a seeded linear probe (50 epochs, learning rate 10^{-2}, seed 42) for the foundation and persona directions; pooling is mean over content tokens. The OLMo-3 decomposition reproduces the prior paper's published projection fraction (0.104 vs. 0.1044), confirming the conventions transferred.

Contrast sets. The refusal direction uses the Arditi et al. harmful/harmless instruction set (Arditi et al., 2024) (400 of each). The moral subspace uses a balanced six-foundation declarative probing set (40 pairs per foundation). The persona direction uses stratified personal/non-personal identity pairs. Behavioral moral judgment uses a 48-scenario moral-foundations probe across four difficulty levels; persona-shift compliance uses borderline requests under four persona framings. The harmful refusal rate in the ablation-layer sweep and strength sweep is measured on a 40-prompt harmful subset, in chat format, classified by an opening-refusal detector; no harmful generations are retained.

Pipeline. Phase 1 (decomposition) extracts, per family, the base moral and persona directions, the instruct persona and refusal directions, and the energy decomposition. Phase 2 (ablation battery) sweeps the ablation layer over depth fractions $\{0.4, 0.5, 0.6, 0.7, 0.8\}$, selects the layer that most reduces the harmful refusal rate, applies the single-direction ablation there with model saving, and runs the comprehension battery on the instruct and ablated models. The Llama controls (random-direction control and ablation-strength sweep) reuse the same ablation operation. The single-direction ablation is $W \leftarrow W - \alpha\, r\, r^\top W$ over attention `o_proj` and MLP `down_proj` at all layers, with $\alpha = 1$ unless swept.

Statistics. Moral-judgment drops are paired per-scenario bootstrap estimates (5000 resamples). The matched-random null perturbs the residual-writing matrices with Gaussian noise of per-matrix Frobenius norm identical to the refusal ablation's; we verified the noise reproduces the ablation's per-matrix removal norms to relative error $\sim 10^{-7}$. The single-vs-full-rank AUC gap uses Ledoit-Wolf shrinkage LDA solved by the Woodbury identity through an $n \times n$ system. Across-band effective rank is computed on the uncentered per-layer refusal directions.

Determinism and compute. Probe initialization and dataset resampling are seeded. Each family's full pass runs on a single A100; per-layer probes are trained on CPU with thread count capped to avoid intra-op thread thrashing, which changes runtime but not numerics. Code and the per-model output JSON (from which every number in this paper is reproducible) are released with the DeepSteer toolkit.